

SELF-POWERED WATCHES THAT YOU NEVER HAVE TO CHARGE

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There are two types of watches: self-powered ones and those powered by batteries or cells. Automatic watches are easy to maintain and durable, as compared to battery-operated watches. These are also precise and accurate.

Self-powered watches are again of three types, depending on their source of power. These include watches powered by body heat, human hand movements and solar power. This article takes you through self-powered watches, their types and working mechanisms.

Smartwatch powered by body heat. Matrix Power Watch X is the world's first power watch that does not

need charging. It is powered by body heat, and can measure calories burned, sleep patterns, steps taken and activity levels using thermoelectric technology. It also displays the levels of electrical power that have been generated during the day. Data is available on the accompanying mobile app. It can be shared with friends to compare who has burnt the most calories. It has high water resistance compared to other smartwatches in the market.

The power watch uses body heat to power itself. When taken off, it saves data in its memory and goes to sleep. When it is put back on, the watch turns itself on right from where it went to sleep. The watch powers itself using components that use less power and a design that best utilises the conversion mechanism of heat into electricity using temperature differences.

It has two versions: Power Watch and Power Watch X. The former costs about ₹ 13,000, while the latter costs around

₹ 20,000. Both have connectivity options supporting Android and iOS. Both come with LED backlights, but only Power Watch X provides notifications.

Automatic or self-winding watch.

This mechanical watch gets power from the natural motion of the wearer. It gets energy from the mainspring, which is wound automatically, making manual winding unnecessary. Mainspring is the spiral spring connected to watch gear. Energy can also be stored manually by turning the knob. This enables the watch



Fully-automatic watch by Boesl



Matrix Power Watch X displays and analyses sleep patterns (left) and shows number of steps taken, calories burned and distances covered (right)



to run even when not worn. Energy from the mainspring powers the watch till it runs down. Then, it requires a spring to be wound again, automatically or manually by hand.

A watch with a fully-wound mainspring can store enough energy to last two days. Leading manufacturers of automatic watches include Fossil and Seiko.

There is another type of automatic watch called chronograph. Its mechani-